Learning Location, Not Looking: A Placement-Memorization Audit and Ten-Episode Repair at the 30M Scale

Anonymous submission

Abstract

Fixed-placement benchmarks let imitation policies learn *location* instead of *looking*. We document this on LIBERO-Object, whose target start poses are exactly pinned (measured) in 6 of 10 tasks and within $\sim \pm 1$ cm in the rest. Our platform is a 30.2 M-parameter deployed stack carrying *no* language model—an open-vocabulary detector's own text tower supplies every text embedding—the smallest surveyed language-conditioned policy under both size conventions at once (17.2 M trained, 30.2 M deployed; the runners-up differ per convention). No result below is attributed to size. First, an audit finds placement memorization at four layers: a calibrated expert whose offset constant encodes pose (sign-flipping under a software rebuild), an iteration-coupled selection loop (0.700 dev vs. 0.300 held-out), a learned grasp head whose regression rationally prefers proprioception to pixels, and—found last, by swapping the instruction—a place head that memorized a command→location map, emitting the correct basket for the command it trained on and points 13–14 cm away for commands it did not, against a basket that never moves; a head trained on all three commands is correct for all three and its swap cell does not collapse, so this layer too is repaired by data rather than architecture. Second, a repair—ten teacher episodes under ± 4 cm placement teleports plus jitter on a nuisance input—restores *visual* grounding, i.e. which input the head reads (attribution: vision 1.1–1.8 cm, proprioception 0.1 cm), verified by a substitution probe paired with a behavioural randomization protocol: the released head scores 0.400 dev, 0.700 held-out, 0.400 randomized, with both non-dev protocols confirmed at n=50 (35/50 [0.56, 0.81]; 26/50 [0.39, 0.65]). The protocol's twin control cells scope that verification from both sides: on the deployment stack the memorized head scores 1/10 under the repaired head's own shift draws (where it scores 4/10), while on a rebuilt stack whose detector shift disables the repaired head's visual goals (0/10), the shell's search absorbs the same shifts for the proprioception-shortcut head (6/10). The behavioural certificate is therefore joint in (head, stack). Third, the policy class that made this reachable is goal-structured decoding driving an engineered pick-and-place shell; free per-tick regression scored 0/56. All results are simulation, one task and one object in every claimed cell, n=10 per cell and final at that n. A dated addendum carries the multi-object campaign: four objects attempted, a second crossed and twice confirmed on never-scored seeds, and two *not* crossed. Five pre-registered predictions were recorded and falsified and six instruments abandoned at their own calibration gates before a seventh, which needs no ground truth, succeeded and retracted our own proposed mechanism: different objects' prompt chains select the *same* detection from the same frame (median centre distance 0.0001), so deployed role binding is **identity-blind** and the crossing object's 35/50 is not explained by binding the named object. Swapping the instruction collapses the cell (7/10→0/10, p = 0.016), and a 2×2 decomposition localises that entirely to the *command embedding*—prompts from another object cost nothing (p = 1.0). Object selection, approach and grasping thus run with no language input at all, and this system's whole language channel is one latched place coordinate: the repaired head is grounded on *a* box, not *the named* one.

1 Introduction

A policy that scores 0.700 on a manipulation benchmark has, on the face of it, learned to see, decide, and act. This paper is about how much of such a number can instead be *location*: constants—in a benchmark's init files, in an expert's calibration, in a learned regressor's shortcut—that encode where the object always is, so that vision only ever gates an approach a memorized value finishes. On LIBERO-Object [11] we measure the preconditions exactly: the target object's start pose is bit-identical across all 50 shipped init states in 6 of 10 tasks, and varies by only 0.25–0.58 cm (std) in the other four. A benchmark with $\lesssim$1 cm of target variance admits position-memorizers, and we found them in every layer of our own system.

The vehicle is MicroVLA, a deliberately small language-conditioned vision-language-action stack: $\sim$30M parameters deployed, with a frozen open-vocabulary detector serving as the only vision *and* text encoder. The size is the paper's setting, not its claim. Its methodological value is that a stack this small has no capacity to paper over systems defects, so every seam had to be instrumented—and the instruments are what this paper contributes.

Contributions.

1. **A placement-memorization audit and its verification method.** A forensically worked instance of shortcut learning [6, 7] on a fixed-placement benchmark, shown at four layers, with the corrected measured account of what LIBERO-Object pins (per-task table and shipped measurement script); and the probe-plus-behavioural-randomization protocol, motivated by the on-manifold limit of substitution probes and by a counterexample to probe-only certification. The protocol's twin control cells sharpen its scope: behavioural randomization separates memorized from repaired heads under identical draws on the deployment stack (1/10 vs. 4/10), and fails to separate them on a stack whose rebuild disables visual goals (6/10 vs. 0/10).

2. **A controlled policy-class result.** With perception, trunk, and corpus regime fixed, free per-tick regression scores 0/56 (upper 95% bound 6.4%) while goal-space supervision driving an engineered pick-and-place shell scores 4/10 dev, 7/10 held-out, 4/10 randomized—claimed strictly as a bundle, with the shell's share measured on both stacks.

3. **Training–serving-skew instrumentation.** 29 worked defect instances at producer/consumer seams, split into disagreements catchable by parity testing and agreements-on-a-wrong-convention catchable only by provenance [3, 16].

4. **A ground-truth-free test for identity-blind grounding, and the negative result it produced.** Open-vocabulary detectors are typically prompted with a fallback chain, so a policy can appear language-conditioned while every instruction collapses to the same generic prompt. Comparing two *different* instructions' selections on the same frame detects this without knowing which is correct—the property that let it succeed where six ground-truth-seeking instruments failed. Applied to our own stack it returns the same box for different objects (median centre distance 0.0001), so our best cell's 0.700 is *not* explained by binding the named object. Its behavioural companion is equally cheap: **swap the instruction to a different object and rescore against the original task**. Neither needs annotation, both need only a second forward pass, and they fail differently—the first exposes a grounding stage that ignores the instruction, the second a stage that memorized it. The swap is what found our fourth memorization layer, and it works precisely where a fixed benchmark cannot: when every task shares a target, a memorized command$\rightarrow$location map and a grounded one are behaviourally identical until the command is changed. Both ship as `eval/probes.py` with a test suite, and their verdicts refuse the three over-claims this campaign got burned by: too few co-detected frames returns INCONCLUSIVE rather than a rate; a discriminating result states that it cannot see whether the discrimination is *correct*; and an underpowered cell prints its own ceiling.

Both headline findings have one shape: apparent competence resting on a channel other than the claimed one—position where we claimed looking, shape where we claimed naming. The audit generalizes better than the artifact does.

2 Platform

No separate language model exists anywhere in the stack: the command is parsed into source/target phrases, and CLIP embeddings for (command, source, target) are harvested once per task from the detector's own text tower. **The world model is causally inert in every nonzero number of this paper**—a persistence-TRM ablation reproduces the held-out cell exactly (0.700).

Table 1: Deployed parameter ledger. "Deployed" counts every parameter loaded at inference, frozen or trained.

component	params	status
YOLO-World-S detector (vision *and* the only text encoder)	13.0 M	frozen
Tiny Recursive Model world model	9.97 M	frozen at deployment
fusion + drift + relational + planner trunk heads	7.0 M	trained ($\leq$9 M cap)
goal heads (grasp 0.17 M + place 0.07 M) + learned gates	0.24 M	trained
deployed total	**30.2 M**	

3 Results

Table 2: Protocol × head. All cells $k/10$ with Wilson 95% intervals; task 0, wrist camera. Audit-stack cells are never pooled with deployment-stack cells (§ stack finding).

protocol	memorized head	**flagship (released)**	sibling A	sibling B
dev (states 0–9)	7/10	4/10	7/10	7/10
held-out (states 10–19)	3/10	**7/10**	7/10	6/10
randomized ±4 cm (same draws)	**1/10**	**4/10**	5/10	3/10
randomized ±4 cm (audit stack)	6/10	0/10	—	—

Confirmed at power: held-out **35/50** = 0.700 $[0.56, 0.81]$; randomized **26/50** = 0.520 $[0.39, 0.65]$. Across a detector-stack rebuild the repaired head reproduces in all three protocols while the memorized head's dev cell collapses 7/10 → 2/10 to its held-out floor: memorization is stack-coupled, *visual* grounding—which input the head reads—is what survived.

4 The multi-object addendum: five falsified predictions, one mechanism retracted, and a fourth memorization layer

A second object (butter) crosses: 10/10 under an early-latch configuration, 6/10 with soup at 4/10 under a single anchor-band configuration with no per-object constants, and two independent pre-registered fresh-seed confirmations (5/10 + 3/10; 5/10 + 4/10 on a three-object head). Two further objects (cream cheese, chocolate pudding) do not cross. Goals are accurate where measurable (cream 1.58 cm on-manifold, vs. 1.34 and 1.06 for the objects that work) while each failing teacher never leaves its visual-servo phase, at source-uv std ∼0.30, settling tens of centimetres from the target. Four objects were attempted; two crossed.

A proposed mechanism, tested and retracted. We previously located the failure at *role binding*: the region-text head scores 0.000 on every product name these tasks are written in (measured, and correct), so groceries fall back to a shared generic tail, which would make same-shaped objects indiscriminable by construction. Screening each chain element per object refutes this as a *cause*: soup resolves to "box" at 0.96, butter 1.00, cream 0.92—**the objects that cross fall back exactly as the one that fails does**, and soup succeeds 35/50 while indiscriminable in the stated sense. A companion pre-registered prediction, that box-centre spread would separate them, was falsified in the same run (soup, the best object, has the *highest* spread) and is confounded by a moving wrist camera; abandoned, not retuned. The 0.000 observation stands; the causal story built on it does not.

What replaces it: deployed role binding is identity-blind. The scenes make this decidable without ground truth—soup's contains cream cheese and cream's contains alphabet soup, six of seven objects shared—so rather than ask *where the true object is* (the question that defeated six instruments), we ask whether two objects' chains select the *same detection from the same frame*. They do: same-box rate 1.00 on soup's frames, 0.87–0.96 on cream's, 0.92–1.00 on butter's, median centre distance **0.0001**—200× inside the 0.02 threshold, so threshold-insensitive. Deployed binding carries no object identity, and **soup's 35/50 is not explained by binding the named object**. This fits cream's machine converging within 7 cm of *an* object but not the commanded one, and per-object prompt engineering changing nothing (0/10). It does *not* excuse the four binders: 3.1–3.9 class-agnostic proposals exist per frame ($\geq$2 on 0.71–0.96 of frames, cream highest), so they are handed identity-bearing candidates and fail on their own live scoring—which re-opens that sub-study rather than closing it.

The architecture closes the argument. Code inspection makes this structural rather than statistical: `GraspPointHead.forward` consumes only `geom`, `box_emb`, `frame_emb`, `eef_xy`, assembled by `build_grasp_features(uv, conf, proprio, box_emb, frame_emb)`—no task, command, or text embedding appears anywhere in its signature. The head that decides *where to grasp* therefore receives no language input at all, the only instruction-to-grasp path is *which box is selected*, and that selection is identity-blind. The first two steps hold with certainty and only the third is empirical, so the architecture *predicts* the swap result rather than merely agreeing with it. `PlaceHead` does take a command embedding and is genuinely language-conditioned. Every task in this suite places into the same basket, so all *correct* instructions imply the same place target—which leaves the stack's one real language channel normally *unexercised*, not inert. The swap below exercises it.

Identity-blind is not language-blind. Swapping the instruction to a different object—environment, physics and success criterion unchanged, so success is still scored on the real task—collapses the cell from 7/10 to **0/10** (told butter; 0/10 again told cream cheese; seven discordant pairs one-way, exact two-sided $p = 0.016$) — so the policy plainly does not ignore its instruction. Telemetry places the butter cell's failure *downstream of object approach*: the machine still reaches the true soup object as closely as baseline while `grip_close_rate` falls from $\sim$0.67 to $\sim$0.26.

Decomposition: the collapse is entirely one channel. One instruction drives both the detection prompts and the place head's command embedding, so we drove them from *different* instructions. The four cells form a 2×2: with embeddings from the real task, prompts from another object cost nothing (6/10 vs. 7/10, exact $p = 1.0$); with prompts from the real task, embeddings from another object collapse it (0/10, $p = 0.016$) and reproduce the full swap's per-trial pattern *exactly* (10/10 agreement). One main effect, no interaction. Combined with the architecture, this says what the system is: **object selection, approach and grasping run with no language input at all, and the entire language channel is a single latched** (x, y) **place point** cached once per episode by `set_place(place_head(command_emb))`. Ask this policy for the butter and it finds, approaches and grasps the alphabet soup at baseline rate; it fails only when that one cached coordinate is wrong.

And that coordinate is memorized—the audit's fourth layer. The basket does not move (0.22–0.40 cm across all ten tasks, less than its own 0.8 cm within-task jitter), so a *grounded* place head would emit one point per scene. The released head emits the correct basket for the command it trained on (0.36 cm error) and points **14.5 cm** and **13.0 cm** away for the two it did not: it learned command$\rightarrow$location, not basket—and 14 cm is exactly enough to release a grocery over open table, which is the observed failure. The cause is training coverage, not architecture, and this **repairs** rather than merely explains: on a head trained on all three commands the place-point spread collapses from 14.15 cm to 0.78 cm, and its swap cell *does not collapse*—3/10 against its own 4/10 baseline, exact $p = 1.0$, versus the flagship's 7/10$\rightarrow$0/10 at $p = 0.016$. Same architecture, same manipulation, opposite outcomes; behaviour tracks the place-point measurement exactly, 14 cm destroying the cell and 0.8 cm doing nothing. It also answers generalization: the mechanism reproduces on a second task with a different head, in both directions. This layer was invisible to every earlier probe because all ten tasks share one basket, so a memorized and a grounded place map are behaviourally identical *until the instruction is swapped*.

This scopes the word "grounding". The memorization audit and the vision-vs-proprioception attribution are untouched—they concern which *inputs* the grasp head uses and are measured independently—but the repaired head is grounded on *a box in the image*, not on *the named object*. "Object-level generalization" was consequently never the right frame for this campaign: the pipeline has no object-level channel to generalize over. A system that succeeds without discriminating its target is a weaker claim than one that succeeds by discriminating it, and we report the weaker true one.

Role-binding sub-study (four binders $\times$ three objects, n=10 each, one head and one configuration): the blocked object scores 0/10 under *every* binder—none, crop-CLIP rerank, mean prototype (offline identity 0.613), 1-NN bank (0.902)—while the two that cross move non-monotonically with offline accuracy (butter 4/7/2/4, soup 5/5/8/6). Full cells with run ids are in the manuscript appendix.

Offline binder accuracy dissociates from deployed success. The weaker binder owns the best soup cell and the worst butter cell; the binder with no offline number wins butter. A matched-episode A/B probe explains it: with the 0.902 binder active the uv stream feeding the grasp head *destabilises* (std 0.258 $\rightarrow$ 0.335

lateral, 0.047 → 0.202 vertical)—accurate on corpus crops, thrashing on live ones. The paper's on-manifold limit again: a binder *built* from teacher trajectories cannot bind off-manifold crops.

Five predictions were pre-registered here and all five are recorded as falsified: that 0.82 per-tick identity accuracy would yield 0.914 median-of-3 latch correctness and hence cream > 0 under bank binding (the cell returned 0/10); that chocolate pudding would cross because it rarely won a misbind (its teacher never reached a grasp in five calibration iterations); the box-centre-spread prediction above; that candidate proposals would be too scarce for a binder to choose among (3.1–3.9 exist per frame); and that the instruction swap would be inert (it collapses the cell). Quantifying the deployed side of this boundary was attempted six times and abandoned six times, each at a pre-registered calibration gate: a projected-origin ground truth (rejected — soup, which succeeds 35/50, scored 0.111), its sign-corrected successor (rejected — the in-frame filter deletes the close-up target by construction), three text-tower discrimination probes (rejected — each failed to detect the working object's own phrase), and the spread metric (rejected — confounded by camera motion, and inverted on the best-performing object). Deployed binding accuracy and text-tower discriminability are therefore reported as **unmeasured**, not estimated. The error is attributable—the prediction assumed offline per-tick accuracy equals deployed per-tick accuracy, which the probe refutes and which we do not measure. The prediction, its falsification, and this attribution are retained rather than removed.

5 Related work

Size is comparable only under a stated convention, and the two in common use disagree about which system is smallest, because published counts routinely exclude a frozen language encoder that inference nonetheless loads. We report both (Table 3). Under *trained parameters* the smallest published alternative is Octo-small's 27 M transformer [15], and MicroVLA's 17.2 M is 1.6× below it; under *deployed* = every parameter loaded at inference it is RT-1's 35 M [4], and MicroVLA's 30.2 M is below that *even after granting RT-1 a free language encoder*—RT-1's count excludes the Universal Sentence Encoder producing its instruction embedding, for which we cite no figure rather than estimate one. MicroVLA is thus smallest under both conventions simultaneously, the strongest form of the claim the evidence supports. The survey was run adversarially, against systems whose stated goal is minimal size: NanoVLA-S is 52 M trainable atop a frozen ResNet18 *and* a frozen BERT-base (∼161 M loaded) [14], and LiteVLA reaches a Raspberry Pi on a 256 M SmolVLM backbone [10]. Both sit above MicroVLA on both axes, and NanoVLA is the convention gap in miniature—a paper named for being nano, reporting 52 M while loading ∼161 M, because the frozen language encoder is not counted.

Table 3: Size under both conventions. † excludes a frozen language encoder the published count omits. Sizes only: capability, task breadth, and protocol are not comparable.

system	trained	deployed	language encoder
RT-2 [5]	55 B	55 B	internal (PaLI-X)
OpenVLA [9]	7 B	7 B	internal (Llama-2)
SmolVLA [18]	450 M	450 M	internal (SmolVLM2)
LiteVLA [10]	LoRA only	∼256 M	internal (SmolVLM-256M)
NanoVLA-S [14]	52 M	161 M	BERT-base, frozen
Octo-small [15]	<u>27 M</u>	138 M	T5-base, 111 M frozen
RT-1 [4]	35 M	<u>≥35 M</u>†	USE, frozen, uncounted
MicroVLA	**17.2 M**	**30.2 M**	*none*—detector's own tower

The claim is bounded and checkable: it ranges over language-conditioned VLA policies with published parameter counts that we surveyed, it is a statement about parameter counts alone, and it is refuted by exhibiting any surveyed-class system below 17.2 M trained or 30.2 M deployed. We cannot rule out unpublished or unsurveyed smaller systems. "Language-conditioned" here names the *reference class* — systems that accept a language instruction — and is not a claim about how much work that instruction does in ours: the decomposition above shows it reduces to one latched place coordinate. A reader who thinks that disqualifies the artifact from the class should read the size row as smaller still, not larger. The mechanism behind the gap is architectural rather than incidental: MicroVLA carries no separate language model at all, because the detector's own text tower supplies every text embedding—so the frozen encoder that the other rows pay for twice (once in memory, once in the convention mismatch) does not exist here. Of MicroVLA's 17.2 M trained parameters only 7.24 M are causally load-bearing, since the 9.97 M world model is inert in every nonzero number of this paper. No number here is comparable to community-protocol LIBERO scores (600 steps, n=10, wrist-only, scripted-expert corpora). Predicting *where* rather than *how* is the Transporter [20] and CLIPort [17] lineage, continued by keypoint interfaces over engineered primitives [8, 12] and, with both levels learned, $\pi_{0.5}$ [2]. Our audit is a worked manipulation instance of causal confusion and copycat shortcuts [1, 6, 19]; what we add is the audited counterexample to probe-only certification, the resulting probe-plus-randomization pairing, and the control cells that scope the pairing itself. Bootstrapping from scripted experts is established [13]; our delta is that the scripted teacher is itself audited, caught encoding placement and detector version, and repaired.

6 Claims, non-claims, limitations

Claimed: the pinning measurement in its scoped form; the released head's cells above; free per-tick regression at 0/56; the de-memorization mechanism at the head level (attribution shift, varied-label validation, and a paired jitter ablation, 0/10 → 7/10 dev, exact McNemar/sign $p \approx 0.016$), scoped by the audit-stack control; 29 worked skew defects; the bounded size result of Table 3 (smallest of the surveyed systems under both the trained and the deployed convention simultaneously); and the identity-blindness of deployed role binding, together with its architectural corollary that no text embedding reaches the grasp head—claimed as a property *of this system*, and offered as a test other detector-grounded policies can run, not as a claim about them. **Not claimed:** any task-success contribution from the world model (causally inert in every nonzero number); end-to-end learned motor control (the shell is engineered); head-level grounding from the randomized column alone, stack-free; object-level generalization beyond the addendum's two crossed objects (four attempted, two not crossed and, after a tested-and-retracted mechanism, *not explained*); any deployed binding-accuracy figure; physical deployment; any size claim beyond Table 3's bounded one (surveyed systems, published counts, parameters only)—neither that smaller systems cannot exist, nor that small size caused any result here. **Limitations:** n=10 per cell with n=50 confirmations on two protocols; ten held-out states reused across five disclosed selection looks; machine constants inherited from a dev-tuned era; substitution probes are on-manifold; ±4 cm randomization sits inside the shell's ±6 cm probe envelope and separates heads only where visual goals are live; the teacher's calibration consumed privileged telemetry offline; single benchmark, single simulator.

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